

Welch Two-Sample T-Test: Gender and Height

Project Cycle 3

This notebook performs the main statistical inference for the selected research question:

Is the mean height different between male and female students?

中文研究問題：

男性與女性學生的平均身高是否不同？

Variables Used

- Group variable: Sex
 - Response variable: Height_m
-

Statistical Method

Because height is a quantitative variable, this project compares the mean height between two independent groups.

The method used in this notebook is:

Welch two-sample t-test

Welch t-test is used because it does not assume equal variances between the two groups.

Hypotheses

Null hypothesis:

$$H_0: \mu_{\text{male}} - \mu_{\text{female}} = 0$$

Alternative hypothesis:

$$H_1: \mu_{\text{male}} - \mu_{\text{female}} \neq 0$$

Significance Level

- $\alpha = 0.05$

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In [2]: from pathlib import Path
import pandas as pd
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt
from IPython.display import display

# =====
# 1. 設定資料路徑
# =====

processed_dir = Path(r"C:\Users\curry\OneDrive\桌面\統計實習\project-cycle-3\data")
output_dir = Path(r"C:\Users\curry\OneDrive\桌面\統計實習\project-cycle-3\outputs")

input_file = processed_dir / "YRBS_2007_cleaned_gender_height.csv"

tables_dir = output_dir / "tables"
figures_dir = output_dir / "figures"
summary_dir = output_dir / "summary"

tables_dir.mkdir(parents=True, exist_ok=True)
figures_dir.mkdir(parents=True, exist_ok=True)
summary_dir.mkdir(parents=True, exist_ok=True)

print("讀取資料:", input_file)
print("輸出資料夾:", output_dir)

# =====
# 2. 讀取清理後資料
# =====

df = pd.read_csv(input_file)

print("資料筆數與欄位數:", df.shape)
display(df.head())

# =====
# 3. 保留需要的欄位
# =====

required_columns = ["Sex", "Height_m"]

for col in required_columns:
    if col not in df.columns:
        raise ValueError(f"資料中缺少必要欄位: {col}")

df = df[required_columns].copy()
df = df.dropna(subset=["Sex", "Height_m"])

df = df[df["Sex"].isin(["Female", "Male"])].copy()
df["Height_m"] = pd.to_numeric(df["Height_m"], errors="coerce")
df = df.dropna(subset=["Height_m"])

print("可用資料筆數:", df.shape[0])
print(df["Sex"].value_counts())

# =====
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# 4. 分成 Male 與 Female 兩組
# =====

male_height = df.loc[df["Sex"] == "Male", "Height_m"]
female_height = df.loc[df["Sex"] == "Female", "Height_m"]

n_male = male_height.count()
n_female = female_height.count()

mean_male = male_height.mean()
mean_female = female_height.mean()

sd_male = male_height.std(ddof=1)
sd_female = female_height.std(ddof=1)

print("Male 樣本數:", n_male)
print("Female 樣本數:", n_female)
print("Male 平均身高:", mean_male)
print("Female 平均身高:", mean_female)

# =====
# 5. 建立群體摘要表
# =====

group_summary = pd.DataFrame({
    "Group": ["Female", "Male"],
    "n": [n_female, n_male],
    "Mean_Height_m": [mean_female, mean_male],
    "SD_Height_m": [sd_female, sd_male],
    "Mean_Height_cm": [mean_female * 100, mean_male * 100],
    "SD_Height_cm": [sd_female * 100, sd_male * 100]
})

group_summary = group_summary.round(4)

print("群體摘要表:")
display(group_summary)

group_summary_file = tables_dir / "gender_height_group_summary.csv"
group_summary.to_csv(group_summary_file, index=False, encoding="utf-8-sig")

print("群體摘要表已輸出到:")
print(group_summary_file)

# =====
# 6. Welch two-sample t-test
# =====

t_stat, p_value = stats.ttest_ind(
    male_height,
    female_height,
    equal_var=False
)

# 平均差異: Male - Female
mean_difference = mean_male - mean_female

# Welch 標準誤
se_difference = np.sqrt((sd_male**2 / n_male) + (sd_female**2 / n_female))
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# Welch 自由度
df_welch = ((sd_male**2 / n_male) + (sd_female**2 / n_female))**2 / (
    ((sd_male**2 / n_male)**2 / (n_male - 1)) +
    ((sd_female**2 / n_female)**2 / (n_female - 1))
)

# 95% 信賴區間
alpha = 0.05
t_critical = stats.t.ppf(1 - alpha / 2, df_welch)

ci_lower = mean_difference - t_critical * se_difference
ci_upper = mean_difference + t_critical * se_difference

print("Welch two-sample t-test 結果:")
print("t statistic:", t_stat)
print("df:", df_welch)
print("p-value:", p_value)
print("Mean difference, Male - Female:", mean_difference)
print("95% CI:", ci_lower, "to", ci_upper)

# =====
# 7. 建立檢定結果表
# =====

test_results = pd.DataFrame({
    "Method": ["Welch two-sample t-test"],
    "Mean_Difference_Male_minus_Female_m": [mean_difference],
    "Mean_Difference_Male_minus_Female_cm": [mean_difference * 100],
    "t_statistic": [t_stat],
    "df": [df_welch],
    "p_value": [p_value],
    "CI_95_Lower_m": [ci_lower],
    "CI_95_Upper_m": [ci_upper],
    "CI_95_Lower_cm": [ci_lower * 100],
    "CI_95_Upper_cm": [ci_upper * 100],
    "Alpha": [alpha]
})

test_results = test_results.round(6)

print("Welch t-test 結果表:")
display(test_results)

test_results_file = tables_dir / "gender_height_welch_test_results.csv"
test_results.to_csv(test_results_file, index=False, encoding="utf-8-sig")

print("Welch t-test 結果表已輸出到:")
print(test_results_file)

# =====
# 8. 畫平均差異 95% 信賴區間圖
# =====

plt.figure(figsize=(8, 5))

plt.errorbar(
    x=[mean_difference * 100],
    y=[1],
    xerr=[((mean_difference - ci_lower) * 100), ((ci_upper - mean_difference)

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        fmt="o",
        capsize=8
    )

    plt.axvline(0, linestyle="--")
    plt.yticks([1], ["Male - Female"])
    plt.xlabel("Mean Difference in Height (cm)")
    plt.title("95% Confidence Interval for Mean Height Difference")
    plt.grid(axis="x", alpha=0.3)

    plt.tight_layout()

    ci_plot_file = figures_dir / "gender_height_ci_plot.png"
    plt.savefig(ci_plot_file, dpi=300, bbox_inches="tight")
    plt.show()

    print("信賴區間圖已輸出到：")
    print(ci_plot_file)

    # =====
    # 8b. t-test 分配曲線圖
    # =====

    # 建立 t 分配曲線的 x 軸範圍
    x = np.linspace(-5, 5, 1000)

    # 根據 Welch 自由度建立 t 分配的 y 值
    y = stats.t.pdf(x, df_welch)

    # t 臨界值
    t_critical_positive = stats.t.ppf(1 - alpha / 2, df_welch)
    t_critical_negative = -t_critical_positive

    plt.figure(figsize=(9, 6))

    # 畫 t 分配曲線
    plt.plot(x, y, label="t distribution")

    # 畫拒絕域
    plt.fill_between(
        x,
        y,
        where=(x <= t_critical_negative),
        alpha=0.3,
        label="Rejection region"
    )

    plt.fill_between(
        x,
        y,
        where=(x >= t_critical_positive),
        alpha=0.3
    )

    # 畫臨界值線
    plt.axvline(t_critical_negative, linestyle="--", label=f"-t critical = {t_critica")
    plt.axvline(t_critical_positive, linestyle="--", label=f"t critical = {t_critica")

    # 如果 t 值太大，會超出圖的範圍，所以用文字標示
    if -5 <= t_stat <= 5:

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plt.axvline(t_stat, linestyle="--", label=f"Observed t = {t_stat:.2f}")
else:
    plt.text(
        1.2,
        max(y) * 0.8,
        f"Observed t = {t_stat:.2f}\nfar outside this range",
        fontsize=11
    )

plt.title("Welch Two-Sample t-Test Distribution")
plt.xlabel("t value")
plt.ylabel("Density")
plt.legend()
plt.grid(alpha=0.3)

plt.tight_layout()

t_distribution_plot_file = figures_dir / "gender_height_t_distribution_curve.png"
plt.savefig(t_distribution_plot_file, dpi=300, bbox_inches="tight")
plt.show()

print("t 檢定分配曲線圖已輸出到：")
print(t_distribution_plot_file)

# =====
# 9. 自動產生文字結論
# =====

female_mean_cm = mean_female * 100
male_mean_cm = mean_male * 100
difference_cm = mean_difference * 100
ci_lower_cm = ci_lower * 100
ci_upper_cm = ci_upper * 100

if p_value < alpha:
    decision = "reject the null hypothesis"
    chinese_decision = "拒絕虛無假設"
    conclusion = "there is a statistically significant difference in mean height"
    chinese_conclusion = "男性與女性學生的平均身高存在統計上顯著差異"
else:
    decision = "fail to reject the null hypothesis"
    chinese_decision = "無法拒絕虛無假設"
    conclusion = "there is not enough evidence to conclude that mean height diff"
    chinese_conclusion = "沒有足夠證據顯示男性與女性學生的平均身高存在統計上顯著差異"

interpretation_text = f"""
Welch Two-Sample T-Test Interpretation

Research question:
Is the mean height different between male and female students?

The cleaned dataset includes {n_female} female students and {n_male} male studen

Female mean height: {female_mean_cm:.2f} cm
Male mean height: {male_mean_cm:.2f} cm

The estimated mean difference, Male - Female, is {difference_cm:.2f} cm.
The 95% confidence interval for the difference is from {ci_lower_cm:.2f} cm to {

Welch two-sample t-test result:

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t = {t_stat:.4f}
df = {df_welch:.4f}
p-value = {p_value:.6f}

At alpha = 0.05, we {decision}.
Therefore, {conclusion}.

中文解釋：

本研究使用 Welch two-sample t-test 檢定男性與女性學生的平均身高是否不同。

資料中女性學生共有 {n_female} 人，男性學生共有 {n_male} 人。
女性學生的平均身高為 {female_mean_cm:.2f} 公分，男性學生的平均身高為 {male_mean_cm:.2f} 公分。

男性與女性的平均身高差異為 {difference_cm:.2f} 公分。
95% 信賴區間為 {ci_lower_cm:.2f} 公分到 {ci_upper_cm:.2f} 公分。

檢定結果顯示：
t = {t_stat:.4f}
df = {df_welch:.4f}
p-value = {p_value:.6f}

在  $\alpha = 0.05$  的顯著水準下，我們{chinese_decision}。
因此，{chinese_conclusion}。
"""

print(interpretation_text)

interpretation_file = summary_dir / "gender_height_welch_interpretation.txt"

with open(interpretation_file, "w", encoding="utf-8") as f:
    f.write(interpretation_text)

print("文字結論已輸出到：")
print(interpretation_file)

```

讀取資料： C:\Users\curry\OneDrive\桌面\統計實習\project-cycle-3\data\data processed\YRBS_2007_cleaned_gender_height.csv

輸出資料夾： C:\Users\curry\OneDrive\桌面\統計實習\project-cycle-3\outputs

資料筆數與欄位數： (13062, 4)

	WhatIsYourSex	Sex	HowTallAreYouWithoutShoesInMeters	Height_m
0	2	Male	1.70	1.70
1	1	Female	1.63	1.63
2	1	Female	1.73	1.73
3	1	Female	1.78	1.78
4	1	Female	1.60	1.60

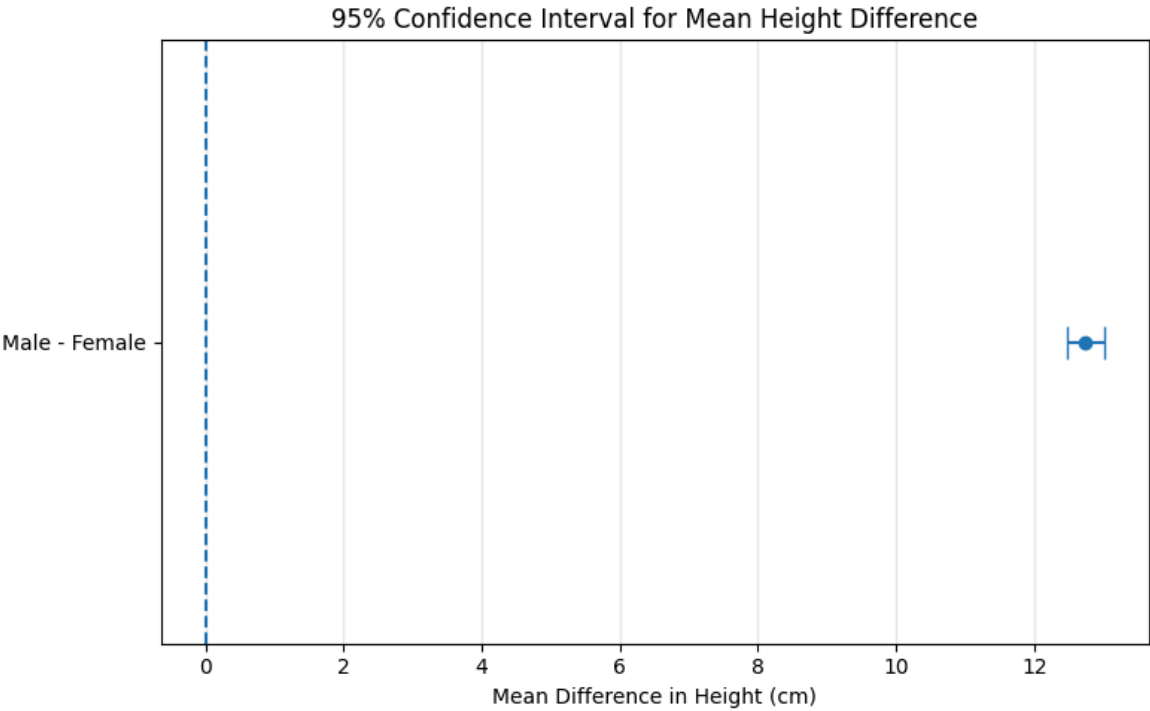
可用資料筆數： 13062
Sex
Male 6572
Female 6490
Name: count, dtype: int64
Male 樣本數： 6572
Female 樣本數： 6490
Male 平均身高： 1.757364576993305
Female 平均身高： 1.62991063174114
群體摘要表：

	Group	n	Mean_Height_m	SD_Height_m	Mean_Height_cm	SD_Height_cm
0	Female	6490	1.6299	0.0738	162.9911	7.3767
1	Male	6572	1.7574	0.0838	175.7365	8.3771

群體摘要表已輸出到：
C:\Users\curry\OneDrive\桌面\統計實習\project-cycle-3\outputs\tables\gender_height_group_summary.csv
Welch two-sample t-test 結果：
t statistic: 92.3128644005698
df: 12892.353675627497
p-value: 0.0
Mean difference, Male - Female: 0.12745394525216502
95% CI: 0.12474762042444928 to 0.13016027007988076
Welch t-test 結果表：

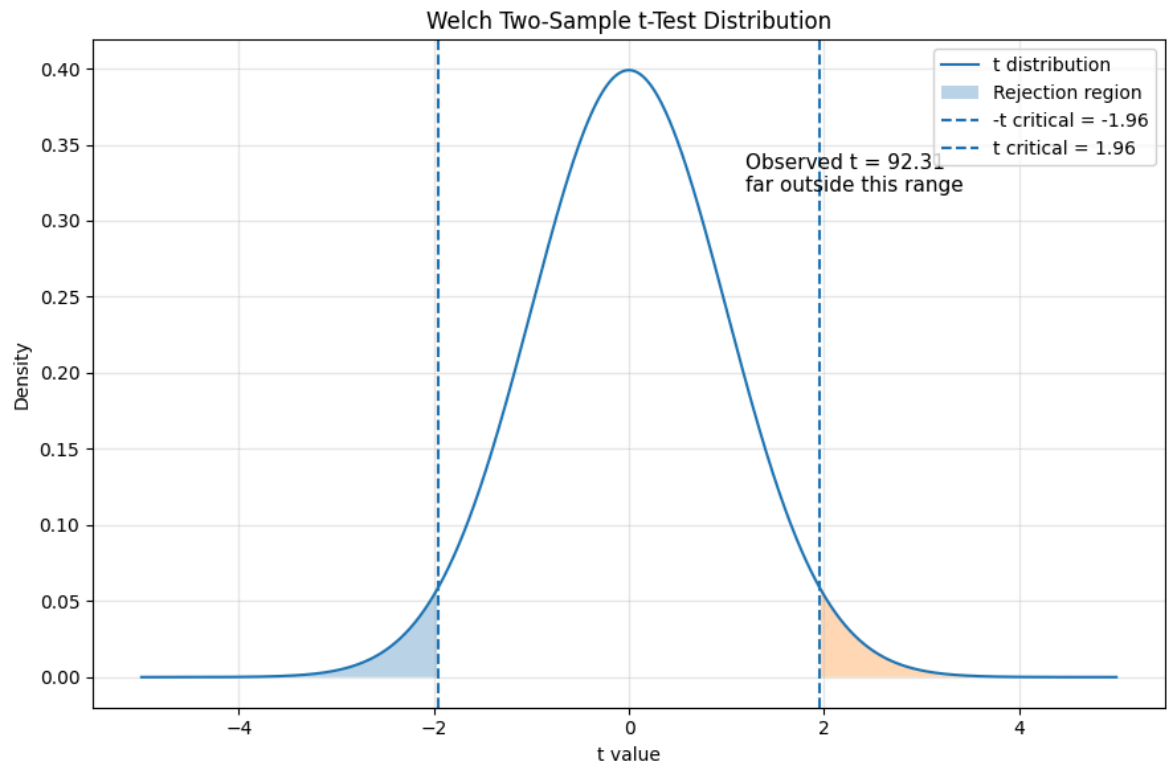
	Method	Mean_Difference_Male_minus_Female_m	Mean_Difference_Male_minus_Fema
0	Welch two-sample t-test	0.127454	12.74

Welch t-test 結果表已輸出到：
C:\Users\curry\OneDrive\桌面\統計實習\project-cycle-3\outputs\tables\gender_height_welch_test_results.csv



信賴區間圖已輸出到：

C:\Users\curry\OneDrive\桌面\統計實習\project-cycle-3\outputs\figures\gender_height_ci_plot.png



t 檢定分配曲線圖已輸出到：

C:\Users\curry\OneDrive\桌面\統計實習\project-cycle-3\outputs\figures\gender_height_t_distribution_curve.png

Welch Two-Sample T-Test Interpretation

Research question:

Is the mean height different between male and female students?

The cleaned dataset includes 6490 female students and 6572 male students.

Female mean height: 162.99 cm

Male mean height: 175.74 cm

The estimated mean difference, Male - Female, is 12.75 cm.

The 95% confidence interval for the difference is from 12.47 cm to 13.02 cm.

Welch two-sample t-test result:

t = 92.3129

df = 12892.3537

p-value = 0.000000

At $\alpha = 0.05$, we reject the null hypothesis.

Therefore, there is a statistically significant difference in mean height between male and female students.

中文解釋：

本研究使用 Welch two-sample t-test 檢定男性與女性學生的平均身高是否不同。

資料中女性學生共有 6490 人，男性學生共有 6572 人。

女性學生的平均身高為 162.99 公分，男性學生的平均身高為 175.74 公分。

男性與女性的平均身高差異為 12.75 公分。

95% 信賴區間為 12.47 公分到 13.02 公分。

檢定結果顯示：

t = 92.3129

df = 12892.3537

p-value = 0.000000

在 $\alpha = 0.05$ 的顯著水準下，我們拒絕虛無假設。

因此，男性與女性學生的平均身高存在統計上顯著差異。

文字結論已輸出到：

C:\Users\curry\OneDrive\桌面\統計實習\project-cycle-3\outputs\summary\gender_height_welch_interpretation.txt